

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Linear Circuit Analysis	Course Code:	EE1001
	Program:	Electrical Engineering	Semester:	Spring 2023
	Duration:	1 Hour	Total Marks:	30
	Exam Date:	10 April 2023	Weight:	15
	Section:	All	Page(s):	2
	Exam Type:	Sessional - 2		

Instruction/Notes:

1. Attempt all questions on the provided answer sheet.
2. Show all steps and box all the final answers with proper units.
3. Attempt all parts of a question together.

CLO 02

Question No. 1

Marks: 10

For the circuit shown below, use node-voltage method to find the power of the dependent source and 1A independent current source.

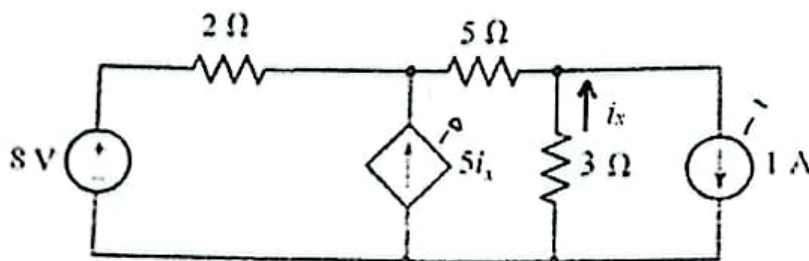


Figure 1. Circuit containing dependent source

Question No. 2

Marks: 10

Use repeated source transformations to find the value of v_x .

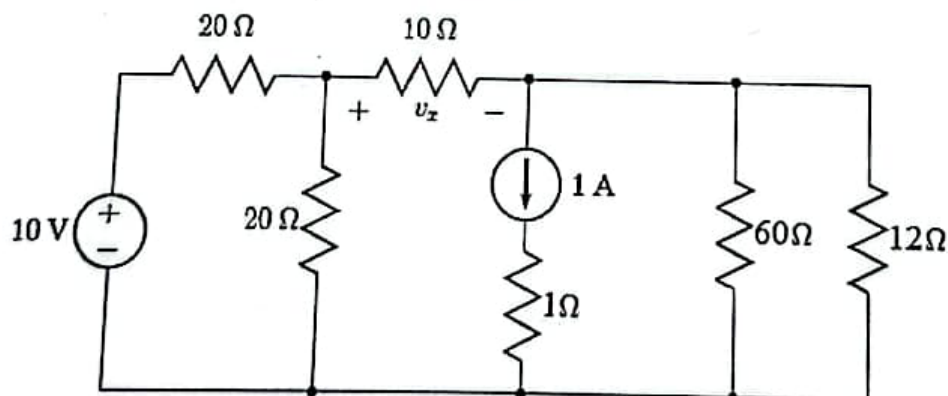


Figure 2. Circuit for Source Transformation

Question No. 3

Marks: 10

For the circuit shown below,

- Calculate the Thévenin equivalent voltage with respect to the terminals a, b using **mesh current method**
- Calculate the Thevenin Equivalent resistance at the terminals a, b using any suitable method.

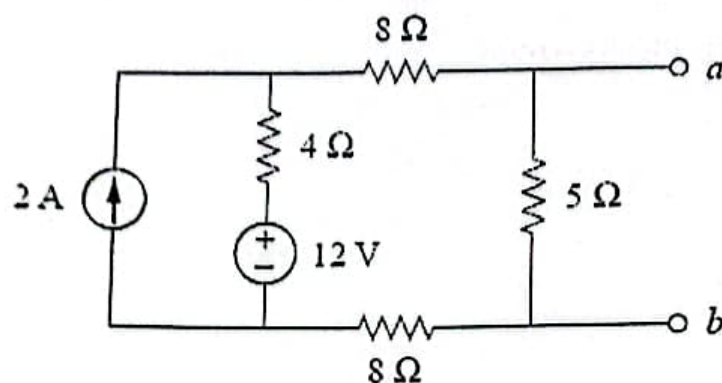


Figure 3. Circuit for finding Thevenin Equivalent